Session Objectives

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1. Recognize and apply **treatment algorithms** for clinical hypertension.

2. Apply the **management** algorithm to specific groups of hypertensive patients.

3. Apply **risk factor modification** in conjunction to medical treatment.

4. Demonstrate a **strategy to differentiate and treat hypertensive emergency and urgency**

Source: Course Syllabus

Acknowledgment

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- Please note this lecture focus on the most comprehensive AHA/ACC 2017 guidelines for HTN
- A new guidelines were released in August 2025 , but we are still in transition and some aspects are still being discussed
- Please note guidelines are recommendations based on systematic reviews of clinical evidence but physicians may still use their clinical judgement when they manage their patients.
- While blood pressure classification stay the same, the 2025 guidelines use the new prevent risk calculator to assess the long term heart disease risk not the ASCVD risk calculator used by 2017 guidelines, (just be aware of both names given we are in transition).
- Despite some differences including the goal, both guidelines aim for early treatment which significantly reduces heart attacks, strokes, and the risk of dementia.
- You may still learn more about the new 2025 HTN guidelines during clinical clerkship when they are more broadly used.

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Let’s review our case from last lecture…

A 65-year-old **black** male with **history of type 2 diabetes mellitus** and **osteoarthritis** presents to the clinic with a **complaint of headaches**. He currently takes **ibuprofen and metformin**. The patient has a **family history of heart failure and hypertension (HTN)**. He lives a **sedentary** lifestyle and **rarely exercises**. His **job is stressful, and he works long hours,** so his diet consists of mostly **fast food**. He has been **smoking** for 35 years and has two **beers** a night. He says he measures his blood pressure at home, and it is **140/80 mmHg**. He brought his cuff in with him today.

Important!

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- Half of all people with hypertension are ***not*** adequately controlled
- Controlling HTN is one of the *most* important components to prevent cardiovascular disease
  - Why? Because it is the most preventable modifiable risk factor for premature cardiovascular disease, more common than smoking, dyslipidemia, or DM!
- Every <u>20 mm Hg</u> increase in systolic BP increases risk of cardiovascular death by <u>2x</u>
- 10 mmHg reduction in office SBP is associated with an approximately 20 percent relative reduction in cardiovascular events

Treatment: Goal Blood Pressure

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- SPRINT Trial: multi-center clinical trial in 2010 to 2015, published in November 2015
  - Enrolled 9361 pts > 50 y/o, most on antihypertensive therapy with a SBP of 130-180 and 1+ risk factors for CVD
  - Standard tx group (targeting SBP <140/<90) vs Control intensive tx group (targeting SBP <120/<90)
  - Key Findings:
    - Intensive tx group significantly reduced cardiovascular events by about 25% and reduced overall risk of death by 27% when compared to standard treatment
  - Intensive treatment was also associated with 4% higher rates of serious adverse effects from anti-hypertensive medications, including syncope, electrolyte abnormalities, acute kidney injury, however, this association was not statistically significant

Treatment: First Line- Lifestyle

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- Lifestyle Modifications
  - Dietary salt restriction
  - Enhanced potassium dietary intake
  - Weight loss in overweight or obese individuals can lead to a significant fall in blood pressure independent of exercise
    - Decline of 0.5 to 2 mmHg for every 1 kg of weight loss
  - The Dietary Approaches to Stop Hypertension (DASH) diet
    - Vegetables, fruits, low-fat dairy, whole grains, poultry, fish, and nuts
    - Low in sweets, sugar-sweetened beverages, and red meats

Q1:

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A 50 y/o M presents to clinic for a routine follow-up. In office, he has a BP of 142/88 mmHg, and upon subsequent measurements he is diagnosed with hypertension. He has no other medical conditions. He does not smoke or drink alcohol. His BMI is 35 kg/m^2. Today, his blood pressure is 144/86 mmHg. He asks about lifestyle changes that can help lower his blood pressure. Which of the following lifestyle interventions has been shown to result in the **greatest** reduction in systolic blood pressure?

A. Engaging in regular aerobic physical activity
B. Reducing dietary sodium intake
C. Following the DASH diet
D. Limiting alcohol intake
E. Weight loss of 20 kgs

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Treatment: Second Line

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- Pharmacologic Therapy
  - When to add?
    - Where lifestyle modifications alone are unsuccessful in lowering BP
    - Atherosclerotic cardiovascular disease 10-year risk = ASCVD10 **(App to calculate: MDCalc)**
      - Age, Diabetes, Sex, Race, Smoker, Total Cholesterol, HDL cholesterol, Systolic blood pressure, Treatment for hypertension
    - Pt has no history of CVD **AND** low risk ASCVD (ASCVD <10%) **AND** SBP ≧140 or DBP ≧90 mmHg
    - Pt has no history of CVD **AND** high risk ASCVD (ASCVD >10%) **AND** SBP ≧ 130 or DBP ≧ 80 mmHg
    - Patient has history of CVD (stroke, MI, etc.) **AND** SBP ≧ 130 or DBP ≧ 80 mmHg

Treatment: Second Line- Pharmacologic Tx

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- Pharmacologic Therapy- lifelong

  - Stage 1 HTN: First line Drugs
    - Thiazide diuretics
    - Calcium Channel Blockers
    - ACE inhibitors
    - Angiotensin II Receptor Blockers

  - Stage 2 HTN:
    - Two first-line drugs of different classes are recommended (one should be ACEI OR ARB, and usually a CCB)

  - ***Thiazides are first line for hypertensive black patients

Treatment: Second Line- Pharmacologic Tx

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Anti-HTN medications	Examples	MOA	Side Effects
Thiazides	Chlorthalidone	Inhibit sodium-chloride transporter in distal tubule	Hypokalemia Decreases urine calcium
Angiotensin Converting Enzyme Blockers (ACE-I)	Lisinopril	Cause vasodilation by inhibiting formation of angiotensin II	Elevated creatinine and hyperkalemia Dry cough Angioedema
Angiotensin Receptor Blockers (ARB)	Losartan	Block angiotensin II (AT_1) receptors on blood vessels and other tissues such as the heart	Elevated creatinine and hyperkalemia
Calcium channel blockers	Amlodipine	Blocks calcium entry into the cell, causing vascular smooth muscle relaxation (vasodilation)	Edema

Other HTN Medications

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Anti-HTN Meds	Example	MOA	Side Effects
Loop Diuretics	Furosemide	inhibit the sodium-potassium-chloride cotransporter in the thick ascending limb	
ENac blockers	Amiloride	directly inhibit sodium channels associated with the aldosterone-sensitive sodium pump	
Aldosterone antagonists	Spironolactone, Eplerenone	antagonize the actions of aldosterone at the distal segment of the distal tubule	Hyperkalemia Spironolactone causes gynecomastia
Beta Blockers	Metoprolol	block the binding of norepinephrine and epinephrine to beta receptors	Bradycardia
Alpha 1 blockers	Terazosin	vasodilation by blocking the binding of norepinephrine to the smooth muscle receptors	Orthostatic hypotension
Direct acting vasodilators	Hydralazine	stimulates formation of nitric oxide by vascular endothelium, leading to cGMP-mediated vasodilation.	Sodium and water retention

Treatment: Second Line- Pharmacologic Tx

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- Pharmacologic Therapy- Special Considerations

Comorbidity + HTN	Medication choice
Heart Failure with Reduced Ejection Fraction (**HFrEF**):	Beta Blocker + ACE-I
Stroke	ACE-I + thiazide
Chronic Kidney Disease	ACE-I/ARB
Diabetes	ACE-I/ARB
Benign Prostatic hyperplasia	Alpha_1 antagonist
Resistant Hypertension	Aldosterone receptor antagonist, beta blocker, direct vasodilator, alpha 2 receptor agonist

- **A diagnosis can be made and treatment started, without further confirmatory readings, in the following uncommon scenarios:**

A patient who presents with hypertensive urgency or emergency (ie, patients with blood pressure ≥180 mmHg systolic or ≥120 mmHg diastolic)

A patient who presents with an initial blood pressure ≥160 mmHg systolic or ≥100 mmHg diastolic and who also has known target end-organ damage (eg, left ventricular hypertrophy [LVH], hypertensive retinopathy, ischemic cardiovascular disease)

Case

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A 50-year-old Black male with no significant past medical history presents to the clinic for an annual check up. He is asymptomatic. His vitals include a BP of 140/88 mmHg, HR 80 bpm, RR 14 breaths per minute, BMI 30 kg/m^2. Physical exam is otherwise unremarkable except for abdominal obesity

What is the next best step in his management?

A. Start medications

B. Bring him in next year to recheck

C. Counsel regarding lifestyle changes and bring in to clinic in 2 weeks to recheck blood pressure

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Case continued

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Patient returned for blood pressure check as advised and blood pressure is 145/90 mmHg. When asked about family history, he mentions that his father was diagnosed with hypertension. If you were to start this patient on a medication, which of the following would be most appropriate for initial monotherapy?

A. Carvedilol

B. Lisinopril

C. Chlorthalidone

D. Hydralazine

E. Clonidine

Case continued

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Case

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A 45-year-old White female with PMHx of HTN and DM type 2 presents to the clinic for follow up regarding hypertension medications. She is currently taking metformin and hydrochlorothiazide. Her blood pressure is 145/90 mmHg. Labs are unremarkable except for mildly elevated creatinine. ECG in the office is consistent with left ventricular hypertrophy. You decide to add another anti-hypertensive medication to her current regimen. Which of the following medications would be appropriate given her history and lab results?

A. Calcium channel blocker

B. Alpha 1 blocker

C. ACE-I

D. Thiazide

E. Aldosterone antagonist

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Summary

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- Exercise and diet (DASH) have a significant impact when it comes to controlling hypertension
- ACE I, ARB, CCB, or Thiazides are first line treatment for hypertension.
- Goal and treatment differ depending on the risk factors and comorbidities of a patient
- Be aware of resistant hypertension and understand that a hypertension specialist should be consulted
- Recognize signs and symptoms that distinguish hypertensive emergency from hypertensive urgency